

Mihir Upadhyay

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EDUCATION

New York University

M.S. in Data Science

GPA: 3.9/4.0

Advisor: Umang Bhatt

Coursework: Independent Research I & II, Deep Learning, Machine Learning, Optimization & Computational Linear Algebra, Probability & Statistics, Responsible Data Science

New York, US

Sep 2024 - Present

Indian Institute of Technology BHU

B.Tech in Chemical Engineering, First Class with Honours

GPA: 8.68/10.0

Coursework: Linear Algebra, Differential Equations, Vector Calculus, Programming in C, Computational Fluid Dynamics, Applications of psychology, New Literature in English

Varanasi, India

Jul 2017 - May 2021

PUBLICATIONS

Characterizing Higher-order Externalities of Human-AI Interaction on Human-Human Interaction *Working paper*
Mihir Upadhyay*, Bernardo Vilegas*, Siobhan Mackenzie Hall, Katherine M. Collins, Umang Bhatt

When Should We Orchestrate Multiple Agents?

Umang Bhatt*, Sanyam Kapoor*, Mihir Upadhyay, Katherine M. Collins, Ilia Sucholutsky, Francesco Quinzan, Adrian Weller, Andrew Gordon Wilson, Muhammad Bilal Zafar

arXiv preprint

Characterizing Web Search in The Age of Generative AI

Elisabeth Kirsten, Jost Grosse Perdekamp, Mihir Upadhyay, Krishna P. Gummadi and Muhammad Bilal Zafar

arXiv preprint

RESEARCH & INDUSTRY EXPERIENCE

TRACE Lab, University of Cambridge

Visiting Research Assistant

Cambridge, UK

Sep 2025 – Present

- Leading the development of open-source framework to systematically characterize the higher order effects of the human-AI interactions on downstream human-human decision making

IBM Thomas J. Watson Research Center

Student Researcher

New York, US

Sep 2025 – Present

- Developing a unified evaluation framework for LLM agents' tool-level schema understanding & task execution ability from multi-hop execution trajectories in multi-agent systems

RC Trust, Ruhr University

Research Intern

Bochum, Germany

May 2025 - Aug 2025

- Designed an evaluation framework for generative search agents across multi-domain corpora, measuring epistemic diversity, temporal robustness, and source sensitivity.
- Utilized advanced topic-modeling and representational analysis methods, including LLOOM concept induction and custom AST-based parsers, to quantify topical coverage and infer latent reference structures.

Flixstock

Technical Product Manager

Gurugram, India

Oct 2022 - Sep 2023

- Built end-to-end generative pipelines combining diffusion models, GANs, semantic segmentation, and vision-language models to enable controllable and high-fidelity image synthesis for e-commerce
- Integrated advanced modules such as ControlNet and LaMa for conditioning and inpainting, improving structural consistency across varied product categories

Standard Chartered Bank

AI/ML Analyst

Bengaluru, India

Jul 2021 - Sep 2022

- Led bank-wide integration of trustworthy ML practices by deploying Truera-based bias, drift, and stability monitoring pipelines for production forecasting models.

Data Science Intern

May 2020 - Jul 2020

- Developed and optimized gradient-boosting and ensemble classifiers on the Credit Card Fraud Detection dataset, improving precision by 5%.

TEACHING

Teaching Assistant

Courant Institute of Mathematical Sciences, NYU

Sep 2025 – Present

New York, US

- Course: Algebra, Trigonometry, and Functions

Teaching Assistant

Center for Data Science, NYU

Jan 2025 – May 2025

New York, US

- Course: Responsible Data Science

SERVICE

Co-organizer

NYU Computational Intelligence, Vision, and Robotics (CILVR) Seminar Series

Nov 2024 – Present

New York, US

Committee Member

NYU Graduate Community Building Group (GCBG) for inclusivity in Data Science

Sep 2024 – May 2025

New York, US

AWARDS

- **Swami Vivekanand Scholarship for Academic Excellence** toward MS funding (\$75K)
- **IIT BHU Merit Scholarship** for the academic years 2017-2021

SKILLS

Programming: Python, Numpy, pandas, scikit-learn, PyTorch, TensorFlow, Hugging Face, LangChain, Transformers, SQL, Shell, HTML, JavaScript, C, C++, Prolific, Docker, AWS

Languages: Braj (native), Hindi (native), English (fluent)